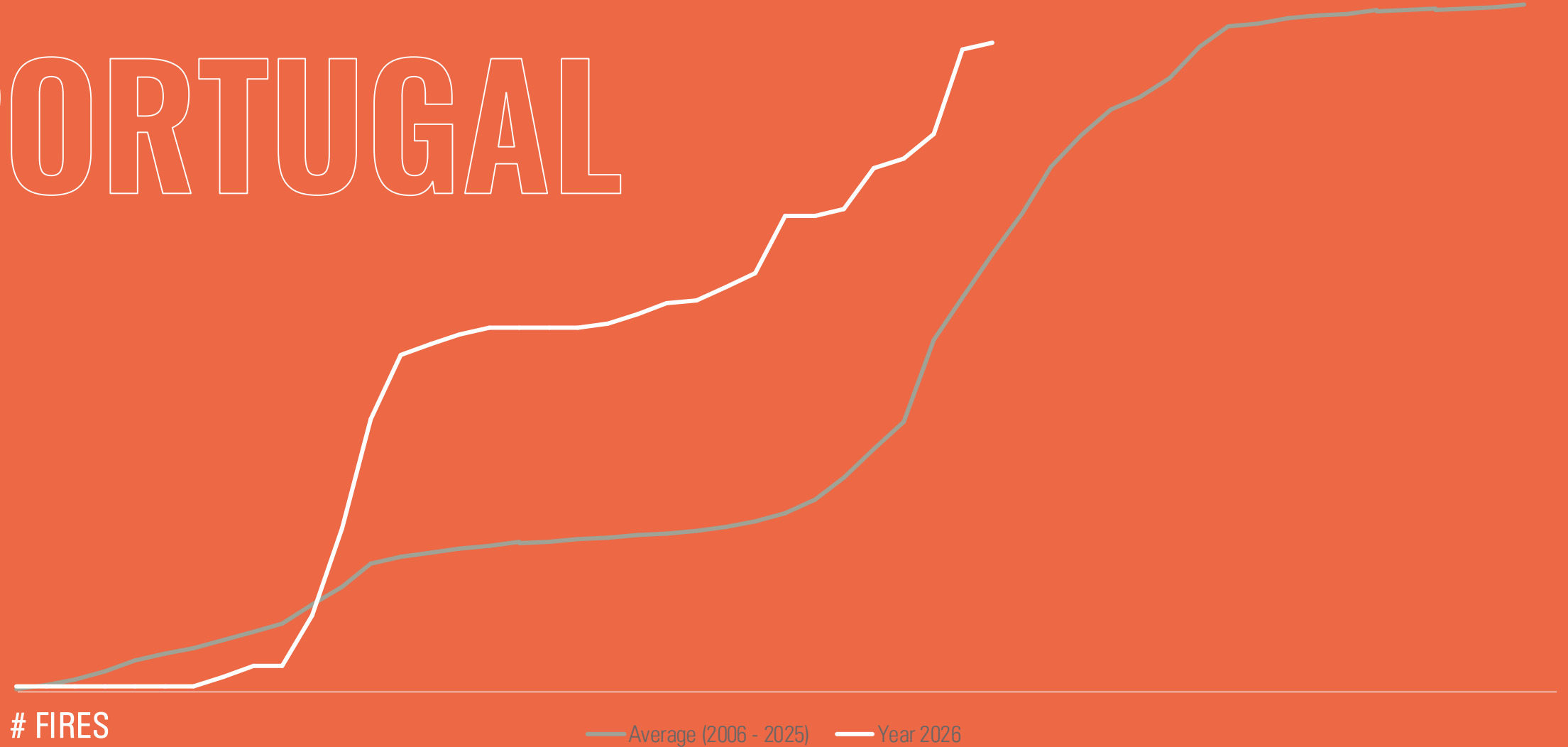


WILDFIRES IN PORTUGAL

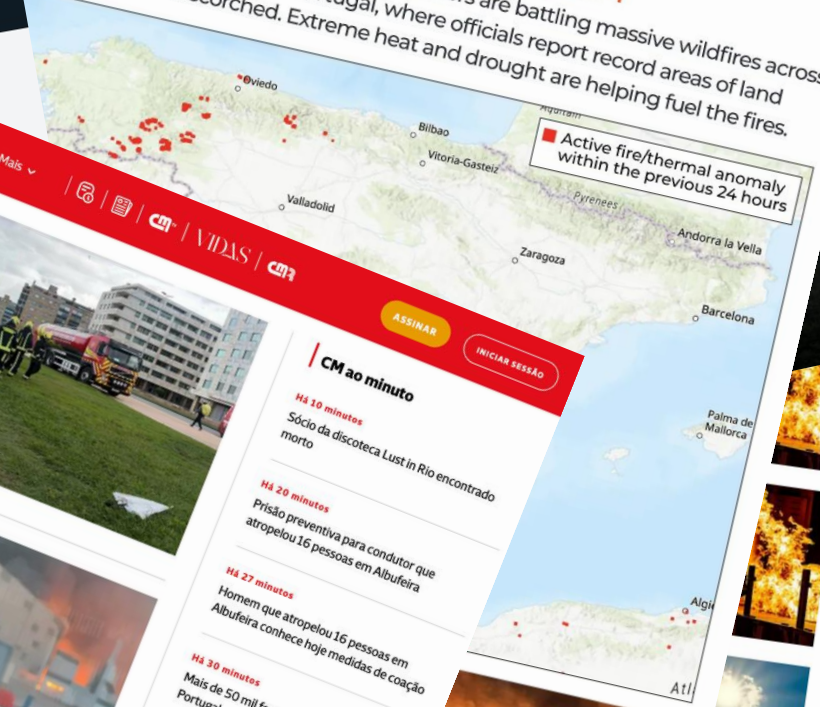




NEWS BUZZ LIFE SPORTS

Wildfires in Portugal and Spain

Thousands of firefighters supported by soldiers are battling massive wildfires across western Spain and northern Portugal, where officials report record areas of land have already been scorched. Extreme heat and drought are helping fuel the fires.



ALERTA: PORTUGAL COM 17 CONCELHOS EM...

NEWS IPMA COLOCA 26 CONCELHOS SOB ALERTA MÁXIMO...

NEWS CALOR DEIXA PAÍS EM ALERTA: HÁ RISCO DE...

NEWS PORTUGAL EM ALERTA PARA OS

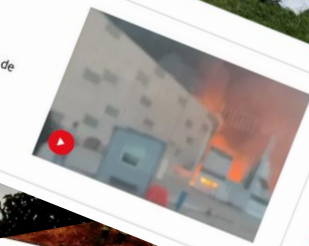
Incêndio provoca danos materiais em bar da praia em Leça da Palmeira

Fogo destruiu a zona da recepção e do arquivo do estabelecimento. 27 de agosto de 2026 às 09:27



Incêndio de grandes dimensões deflagra em armazéns perto de Kiev após ataque russo

Ataque da Rússia, na madrugada desta quinta-feira, fez deflagrar as chamas no distrito de Boryspil. Há também duas casas privadas atingidas. 27 de agosto de 2026 às 07:42



Firefighters battle a forest fire that began near Aldeia de Silva/EPA] Bouses in Galicia, Spain, on August 18, 2025 [Pedro Pascual/Anadolu]



Source: Fire Information for Resource Management System | August 19, 2025



RTP notícias

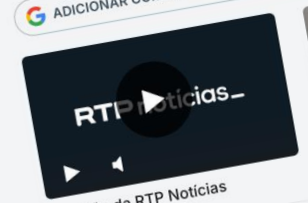
PAÍS

Incêndios em Portugal. A situação

Ao final da tarde de terça-feira, as autoridades descartavam a existência de quatro frentes ativas do fogo. Atualizamos aqui, ao minuto, o estado da situação.

Graça Andrade Ramos, Filipe Alexandre Gonçalves, Andreia Martins, Mariana Ribeiro

ADICIONAR COMO FONTE INFORMATIVA



Emissão da RTP Notícias

MOMENTOS-CHAVE

00:13, 19 Agosto 2026
A situação dos incêndios em Portugal

Source: Copernicus Sentinel-2 imagery, 7 Aug. Firms fire data 8 Aug



Copyright A

Von João Almeida





NEWS BUZZ LIFE SPORTS

CLIMATE

Wildfires in Portugal and Spain

Thousands of firefighters supported by soldiers are battling massive wildfires across western Spain and northern Portugal, where officials report record areas of land have already been scorched. Extreme heat and drought are helping fuel the fires.



ALERTA: PORTUGAL COM 17 CONCELHOS EM...

NEWS
IPMA COLOCA 26 CONCELHOS SOB ALERTA MÁXIMO...

NEWS
CALOR DEIXA PAÍS EM ALERTA: HÁ RISCO DE...

NEWS
PORTUGAL EM ALERTA PARA OS

Nature park Odemira

RTP NOTÍCIAS DESPORTO TELEVISÃO RÁDIO RTP PLAY RTP PALCO RTP ARQUIVOS Entrar

PAÍS EUROPA MUNDO CULTURA OPINIÃO VÍDEOS

Incêndios e

ADICIONAR COMO FONTE

RTP Notícias

MOMENTOS-CHAVE

00:13, 19 Agosto 2026
A situação dos incêndios em Portugal

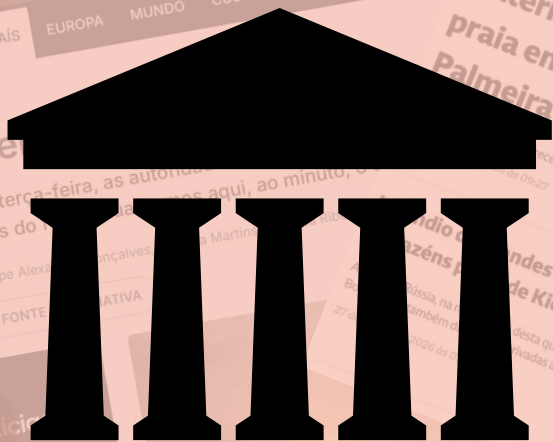
Incêndio provoca danos materiais em bar da praia em Leça da Palmeira

Grandes dimensões deflagram na praia de Leça da Palmeira após ataque russo



Firefighters battle a forest fire that began near Aldeia de Silva/EPA]

August 19, 2025



Article

Spatial Patterns and Intensity of Land Abandonment Drive Wildfire Hazard and Likelihood in Mediterranean Agropastoral Areas

Michèle Salis ^{1,*}, Liliانا Del Giudice ¹, Roghaye Jahdi ^{1,2}, Fermin Alcasena-Urdiroz ³, Carla Scarpa ¹, Grazia Pellizzaro ¹, Valentina Bacci ¹, Matilde Schirru ¹, Andrea Ventura ¹, Marcello Casula ¹, Fabrizio Pedes ¹, Annalisa Canu ¹, Pierpaolo Duce ¹ and Bachisio Arca ¹

¹ National Research Council of Italy, Institute of BioEconomy (CNR IBE), 07100 Sassari, Italy
² Faculty of Agriculture and Natural Resources, University of Mohaghegh Ardabili, Ardabil 56199-11367, Iran
³ Department of Agricultural and Forest Engineering, University of Lleida, 25003 Lleida, Spain
* Correspondence: michele.salis@ibe.cnr.it

Abstract: In Mediterranean agropastoral areas, land abandonment is a key driver of wildfire risk as fuel load and continuity increase. To gain insights into the potential impacts of land abandonment on wildfire risk in fire-prone areas, a fire-spread modeling approach to evaluate the variations in wildfire potential induced by different spatial patterns and percentages of land abandonment was applied. The study was carried out in a 1200 km² agropastoral area located in north-western Sardinia (Italy) mostly covered by herbaceous fuels. We compared nine land abandonment scenarios, which consisted of the control conditions (NA) and eight scenarios obtained by combining four intensity levels (10, 20, 30, 40%) and two spatial patterns of agropastoral land abandonment. The abandonment scenarios hypothesized a variation in dead fuel load and fuel depth within abandoned polygons with respect to the control conditions. For each abandonment scenario, wildfire hazard and likelihood at the landscape scale was assessed by simulating over 17,000 wildfire seasons using the minimum travel time (MTT) fire spread algorithm. Wildfire simulations replicated the weather conditions associated with the largest fires observed in the study area and were run at 40 m resolution, consistent with the input files. Our results highlighted that growing amounts of land abandonment substantially increased burn probability, high flame length probability and fire size at the landscape level. Considering a given percentage of abandonment, the two spatial patterns of abandonment generated spatial variations in wildfire hazard and likelihood, but at the landscape scale the average values were not significantly different. The average annual area burned increased from about 2400 ha of the control conditions to about 3100 ha with 40% land abandonment. The findings of this work demonstrate that a progressive abandonment of agropastoral lands can lead to severe modifications in potential wildfire spread and behavior in Mediterranean areas, thus promoting the likelihood of large and fast-spreading events. Wildfire spread modeling approaches allow us to estimate the potential risks posed by future wildfires to rural communities, ecosystems and anthropic values in the context of land abandonment, and to adopt and optimize smart prevention and planning strategies to mitigate these threats.

Keywords: wildfire simulations; MTT algorithm; land abandonment; wildfire behavior and spread; burn probability; Mediterranean Basin



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1. Introduction

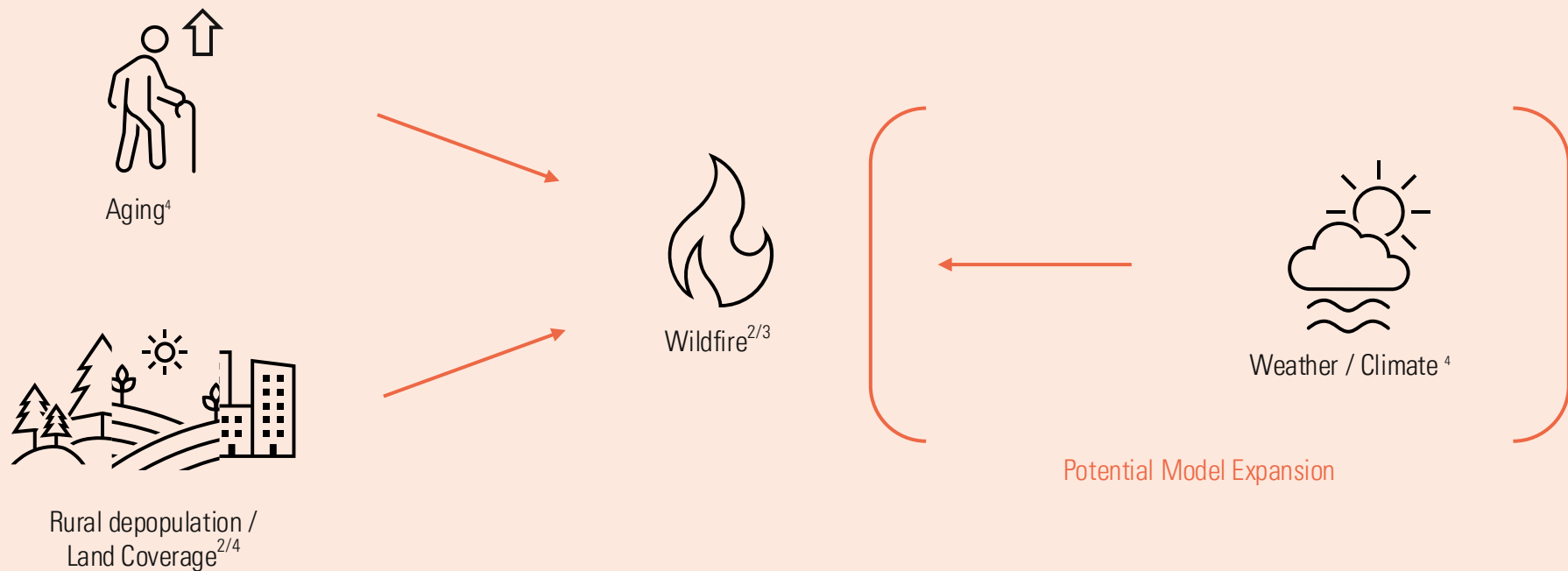
In Mediterranean areas, wildfires are an important disturbance agent and cause significant losses to ecosystems, farms, anthropic values and, sometimes, even human lives [1–5]. During the 2011–2020 period, south-western Europe (Italy, France, Spain, Portugal, and Greece) experienced an average of about 38,000 fire ignitions per year with

Land 2022, 11, 1942. <https://doi.org/10.3390/land11111942>

<https://www.mdpi.com/journal/land>

GOAL

RELATIONSHIP BETWEEN DEMOGRAPHIC CHANGES AND FIRE INCIDENCES?



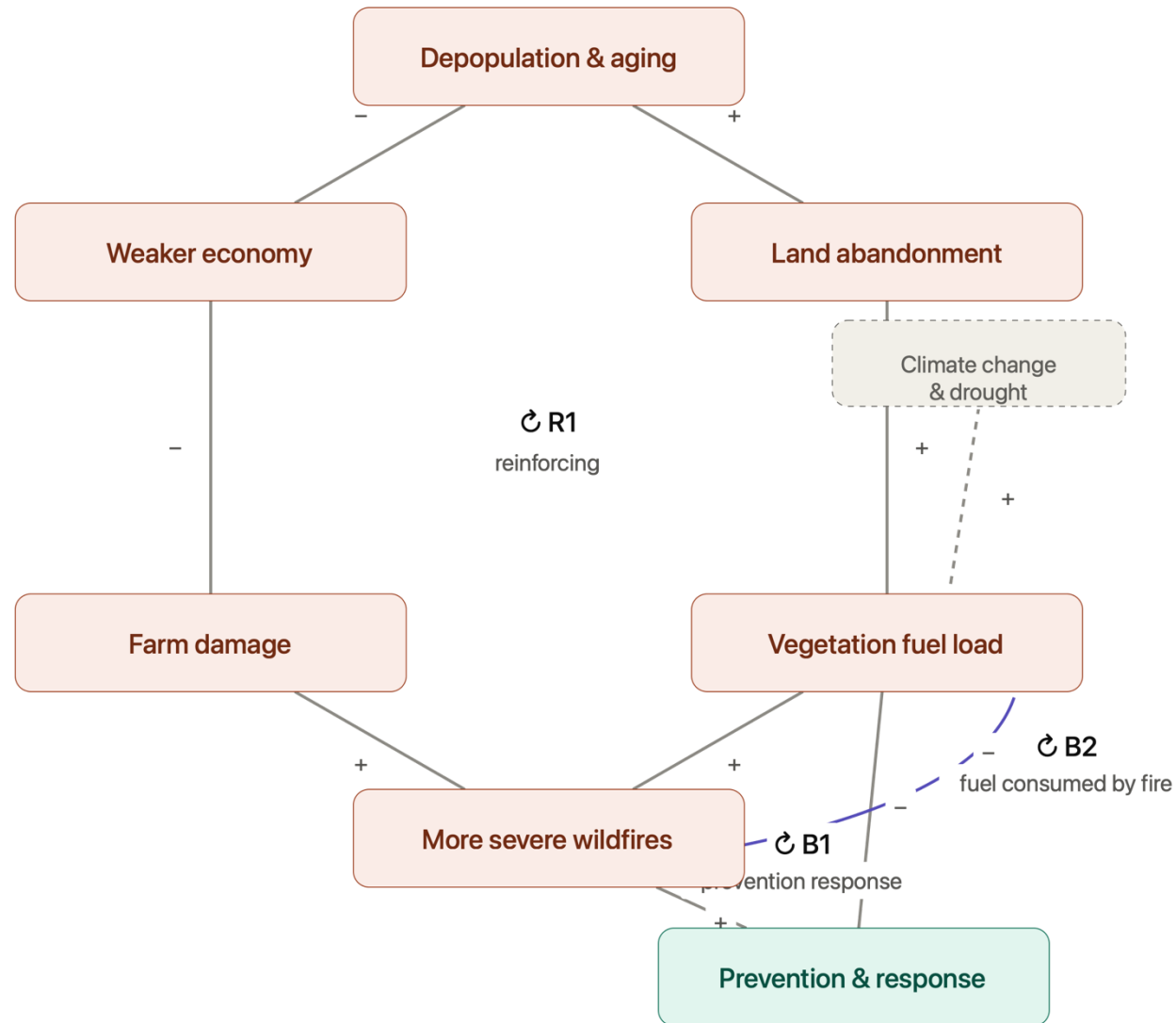
² EFFIS (2026)

³ ICNF (2026)

⁴ INE (2026)

⁵ Climate Copernicus (2026)

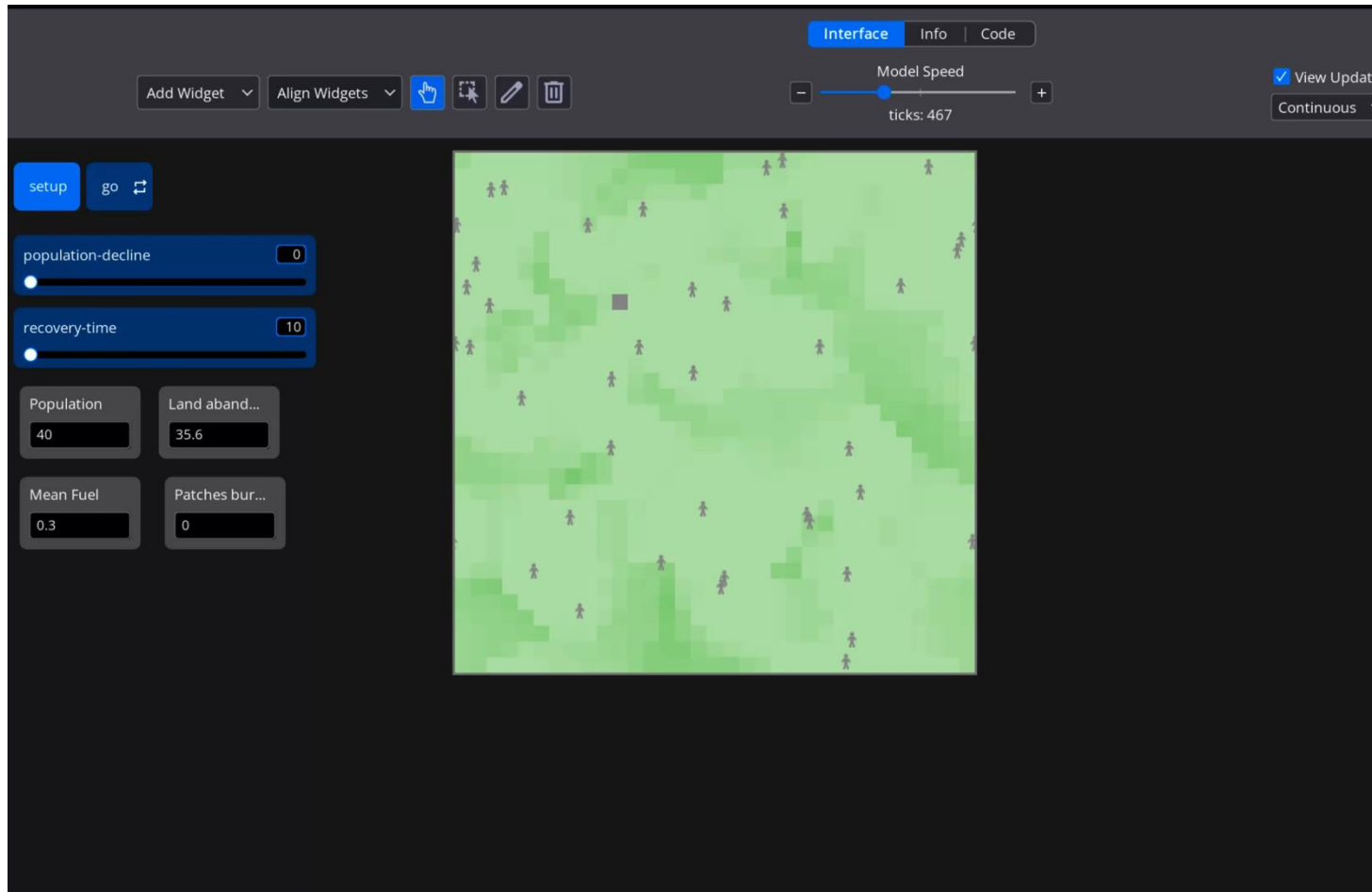
CAUSAL LOOP ¹



R = Reinforcing Loop
B = Balancing Loop

¹ Salis et al. (2022)

AGENT BASED MODELLING

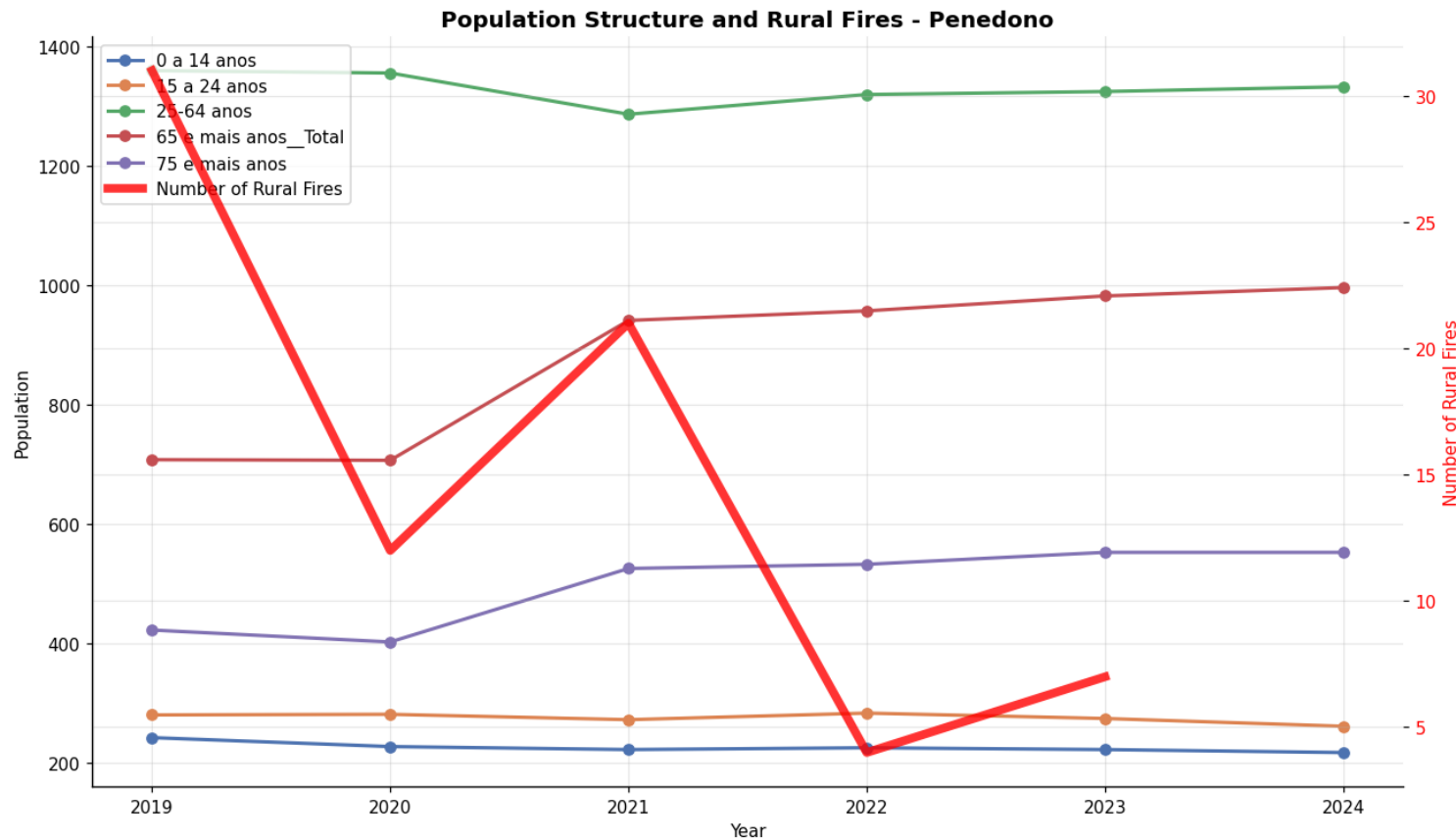


Light green = managed /
low fuel
Dark green = high fuel
Red = burning
Grey = burnt / recovering

PENEDONO – CASE STUDY EXEMPLE



PENEDONO – CASE STUDY EXEMPLE



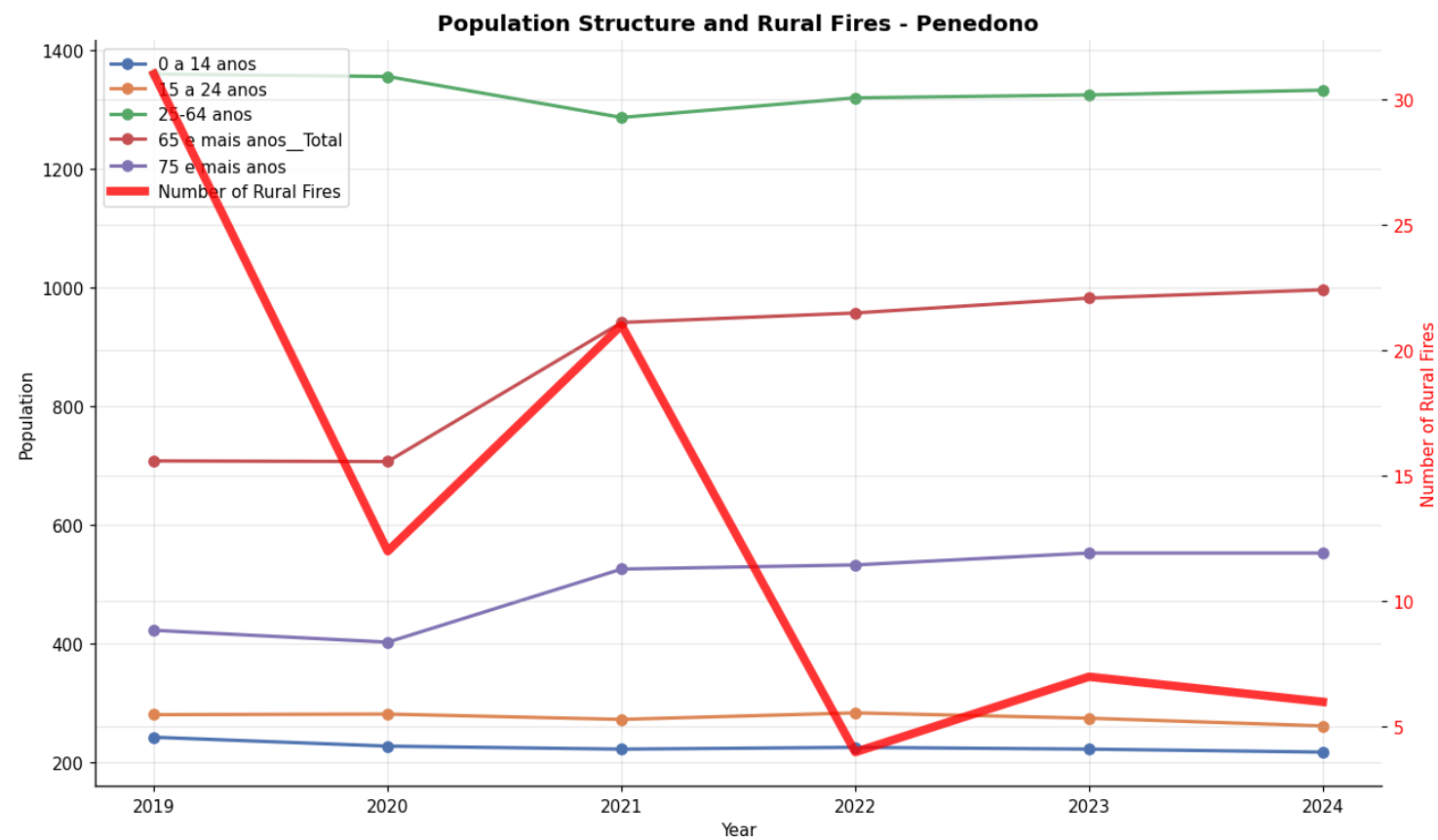
What will happen in 2024?
What is your guess?

Option A: 6 rural fires
Option B: 12 rural fires
Option C: 18 rural fires

³ ICNF (2026)

⁴ INE (2026)

SOLUTION



Option A: 6 rural fires

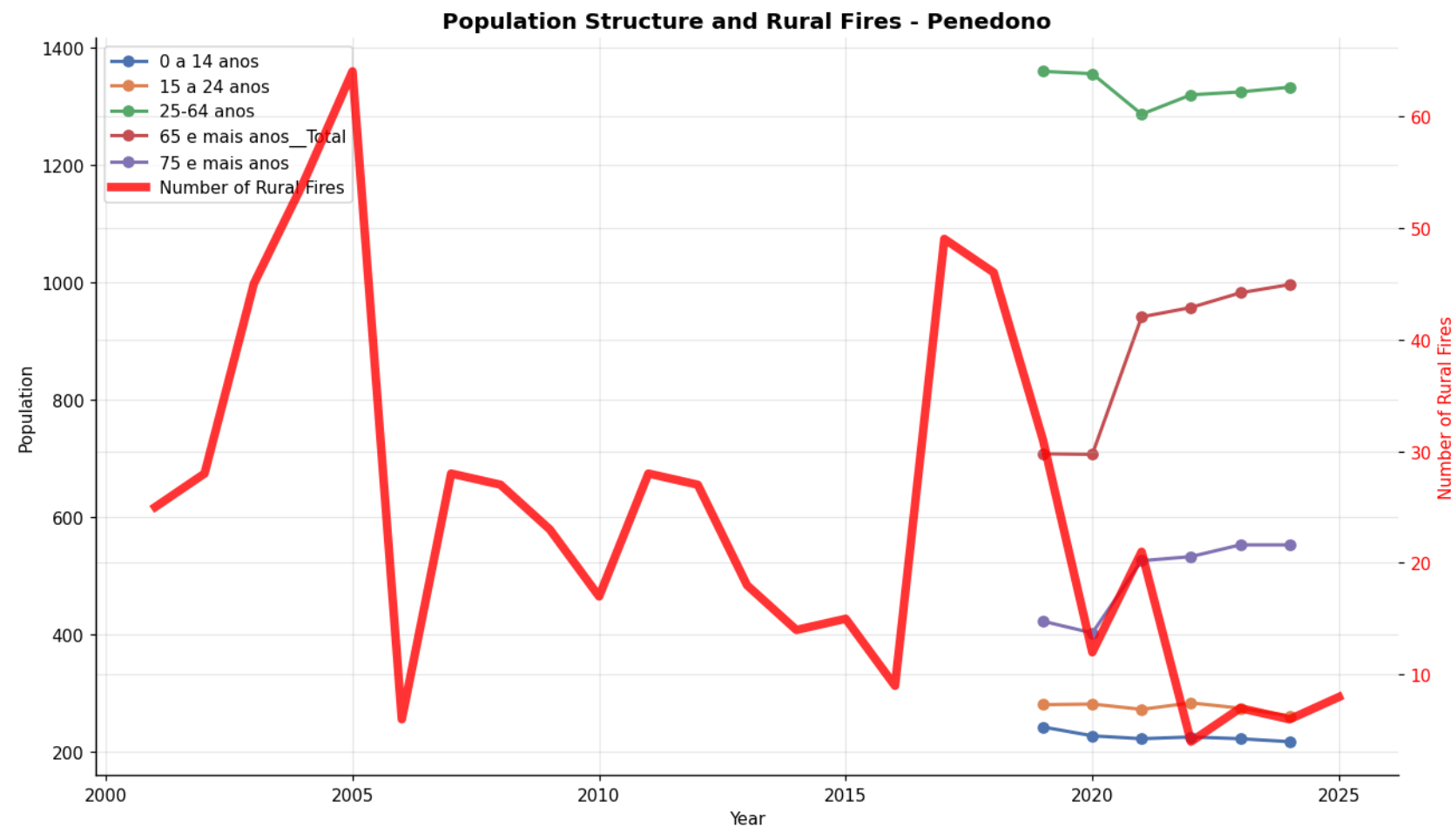
Option B: 12 rural fires

Option C: 18 rural fires

³ ICNF (2026)

⁴ INE (2026)

THE BIG PICTURE



³ ICNF (2026)

⁴ INE (2026)

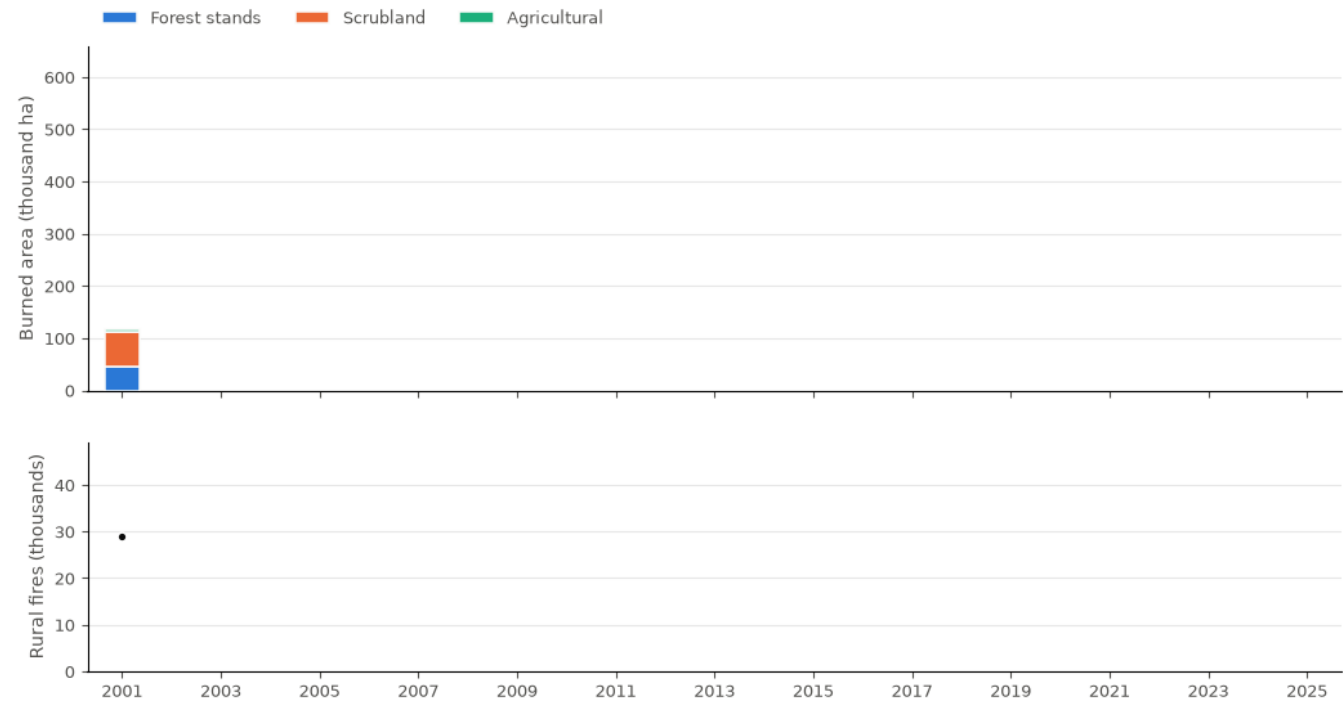
THE REALITY



EDA: SEVERITY IS KEY

Portugal has far fewer fires than in 2001 — but not less burned land

Mainland Portugal, 2001–2025 · ICNF, Estatísticas de Incêndios Rurais (SGIF)



³ ICNF (2026)

OUTLOOK: NEXT STEPS FOR THE ANALYSIS

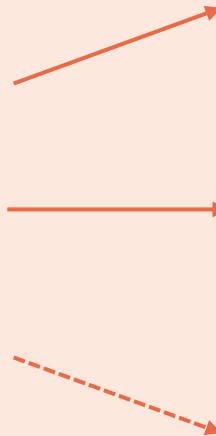
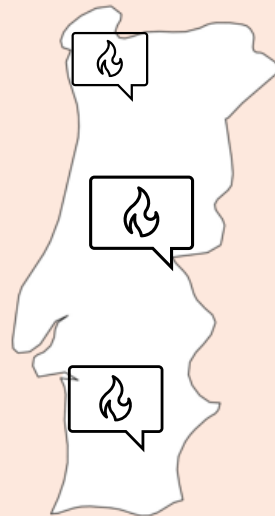
1. Do the trends we expect actually show up in the data?
(Trend analysis)
 2. Does fire risk in a municipality persist, or does it reset after a fire?
(Autocorrelation analysis)
 3. Do population factors impact wildfire risk with a lag?
(Lag & Granger-causality analysis)
 4. Which regions are most at risk, and which factors matter most?
(Predictive risk modeling)
-

LIMITATIONS

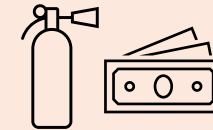
- Data Availability:
 - Full Coverage: 2019 – 2024
 - Periodicity: Yearly
 - Geographical Distance (Municipality level)
 - No Weather / Climate Data included so far
-

VISION

RISK PREDICTION + RESSOURCE ALLOCATION



Risk Prediction



Ressource Allocation

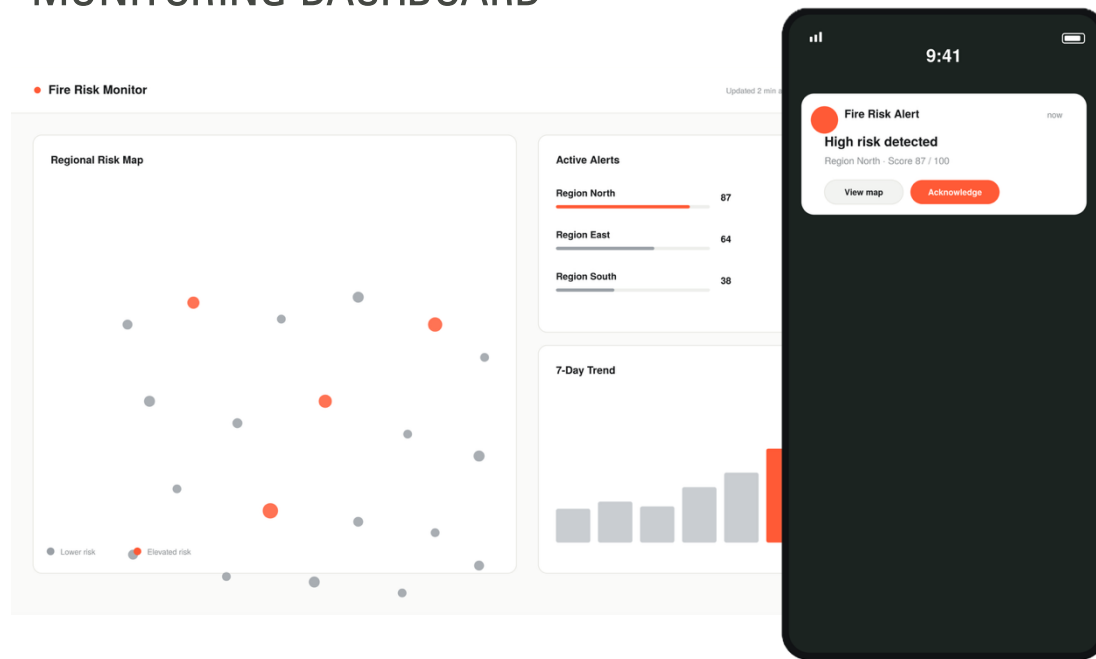


Education /
Awareness

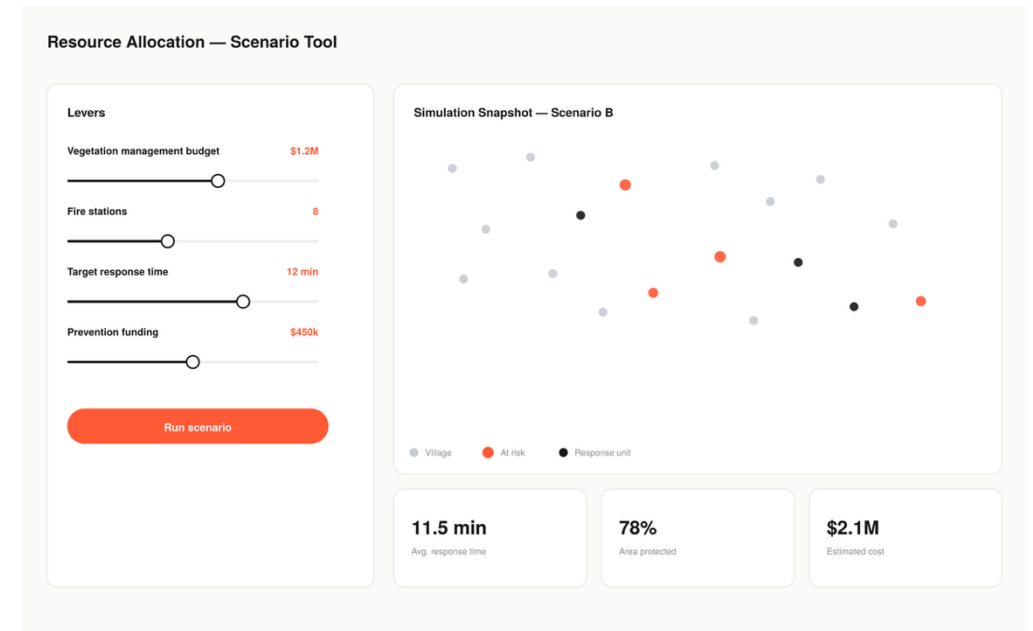
VISION

MONITORING DASHBOARD

+ Alerting for firefighters



SCENARIO-TOOL FOR RESSOURCE ALLOCATION

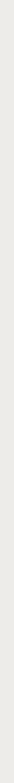


- Rural land management (e.g., select types of vegetation, cleaning private lands mandatory)
- Pre-position firefighting resources in high-risk rural areas (e.g., building water reservatories in high risk regions)

Q+A



APPENDIX



REFERENCES

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-

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- a) <https://www.cmjornal.pt/maissobre/incendio>
- b) https://www.rtp.pt/noticias/pais/incendios-em-portugal-a-situacao-ao-minuto_e1760070
- c) <https://de.euronews.com/2026/05/21/brandgefahr-portugal-sommer-vorbereitungen>
- d) <https://24horas.pt/noticia/portugal-em-alerta-2026-pode-ser-critico-em-incendios>
- e) <https://iowaclimate.org/2023/08/09/shock-news-its-hot-in-portugal/>
- f) <https://www.portugalresident.com/wildfire-warning-its-a-powder-keg-from-north-to-south/>
- g) <https://www.aljazeera.com/gallery/2025/8/19/wildfires-rage-across-spain-and-portugal-as-record-area-of-land-burnt>
- h) <https://exame.com/mundo/as-fotos-do-grave-incendio-florestal-em-portugal/>
- i) <https://www.qatar-tribune.com/article/9804/world/hundreds-dead-in-spanish-and-portuguese-heatwave-report>

Penedeno Pictures [retrieved 27.08.2026]:

- j) <https://www.jn.pt/pais/artigo/empresario-de-sernancelhe-perde-oficina-e-stand-com-prejuizo-de-mais-de-um-milhao/17858881>
 - k) <https://www.portugaltravel.org/castle-of-penedono>
-

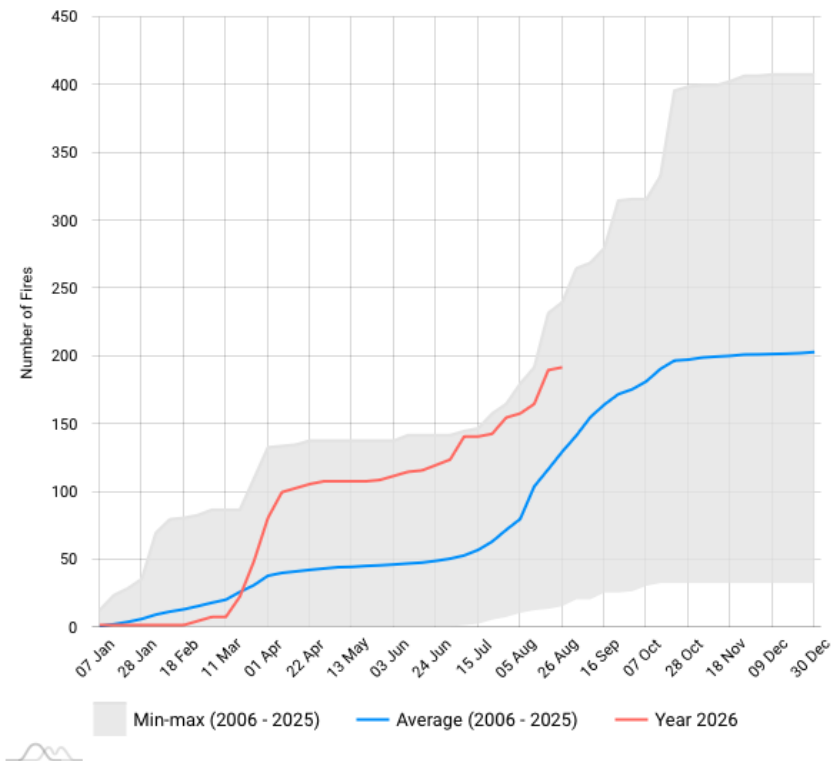
AI DECLARATION

Generative AI was used as a as a supporting tool throughout the development of this project. It assisted with refining and paraphrasing content to improve clarity and professionalism, supporting project planning and providing guidance with coding and technical problem-solving.

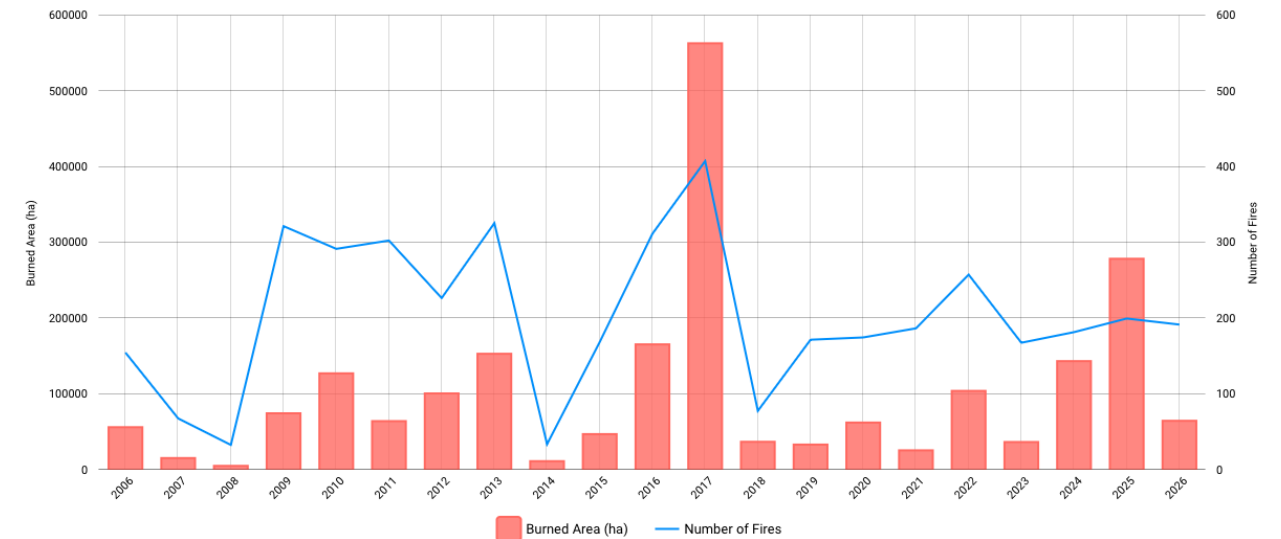


EDA: WILDFIRE IN PORTUGAL

FIRES CURRENT YEAR VS. PREVIOUS⁷



SEVERITY (BURNED AREA) VS. # OF FIRES⁸



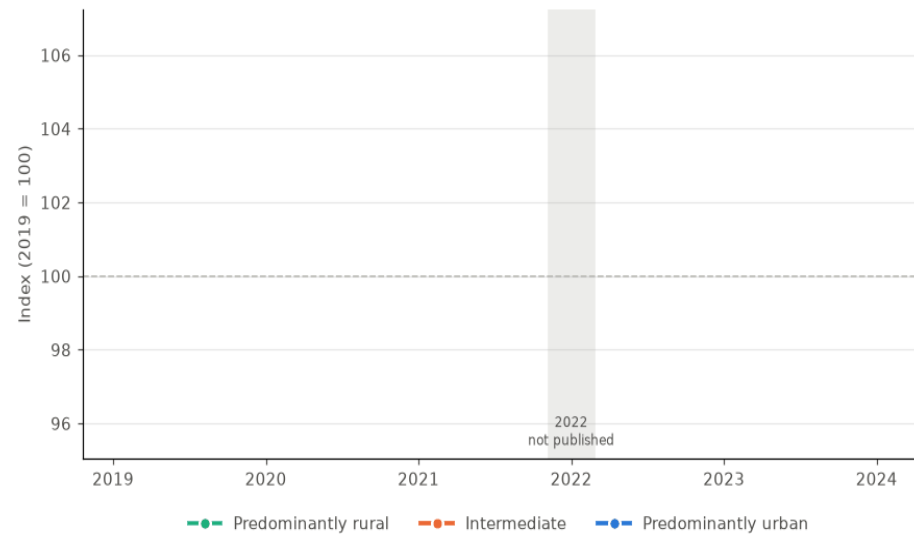
⁷ Copernicus (2026): <https://forest-fire.emergency.copernicus.eu/apps/effis.statistics/seasonaltrend> [retrieved 28.08.2026]

⁸ Copernicus (2026): <https://forest-fire.emergency.copernicus.eu/apps/effis.statistics/estimates/PRT> [retrieved 28.08.2026]

EDA: POPULATION TRENDS

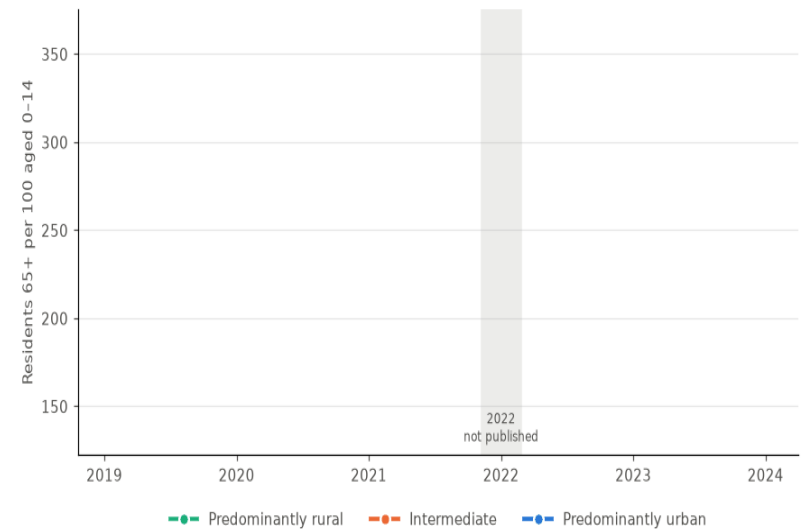
Rural Portugal is losing people while the country grows

Resident population, indexed to 2019 = 100 · INE Anuário Estatístico Regional, table II.1.5



Rural areas are ageing fastest, from an already old base

Ageing index: residents aged 65+ per 100 aged 0-14 · INE Anuário Estatístico Regional, table II.1.5

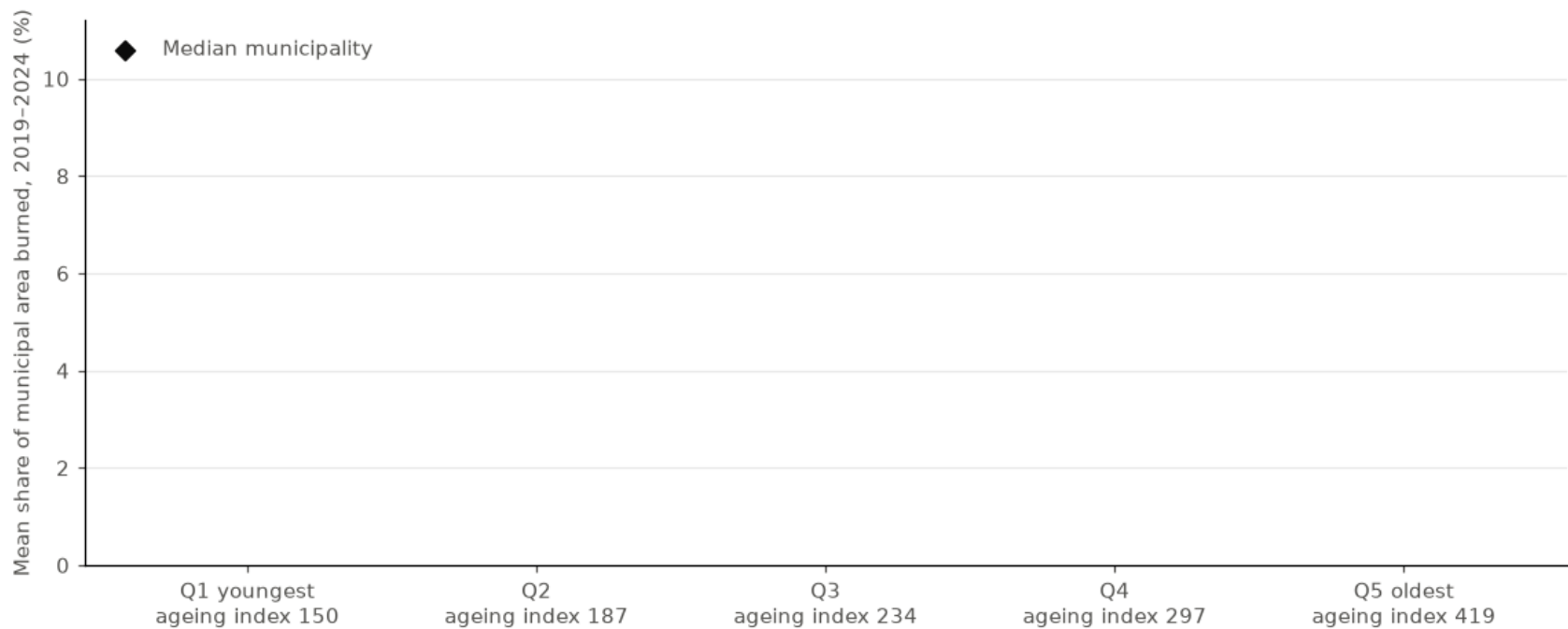


³ ICNF (2026)

EDA: AGING PATTERN

Older municipalities burn more of their own land

Share of municipal land area burned 2019–2024 (EFFIS) by ageing-index quintile (INE) · 278 mainland municipalities



³ ICNF (2026)

SDG CONTRIBUTIONS

